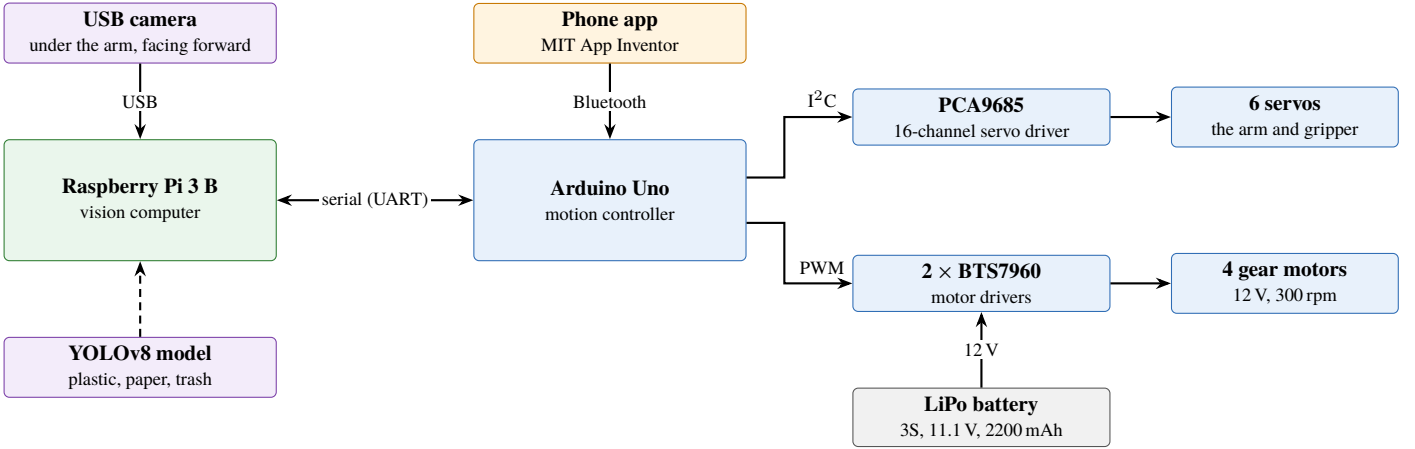
USB camera
under the arm, facing forward
USB
Raspberry Pi 3 B
vision computer
YOLOv8 model
plastic, paper, trash
serial (UART)
Phone app
MIT App Inventor
Bluetooth
Arduino Uno
motion controller
I2C
PCA9685
16-channel servo driver
6 servos
the arm and gripper
PWM
2 × BTS7960
motor drivers
4 gear motors
12 V, 300 rpm
12 V
LiPo battery
3S, 11.1 V, 2200 mAh
computer
motion
vision
control
power